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Exact State Contraction

A definition, preservation theorem, and notation proposal

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Abstract

We define exact state contraction relative to a declared observation contract. The retained transition and observation maps commute with the full system. This identity preserves every finite observation trace by induction. An affine interval extension carries a quadratic accumulated cost through the same associative composition, including cross-terms. We distinguish output preservation from witness-count preservation and specify the printed notation ⊂., spoken TAA. This paper names a contract and gives its elementary preservation theorem; it does not claim priority for quotient systems, bisimulation, or affine composition.

1 The contract

Let $M = (S, U, F, Y, G)$ be an input-driven system. A state $s \in S$ and an input $u \in U$ produce the next state $F(s, u)$ and an observation $G(s, u)$. The observation identifies the property to preserve.

Definition 1 (Exact state contraction)**.** *An exact state contraction of M for G consists of a set Q, a surjection $\pi : S \to Q$, and maps $\bar{F} : Q \times U \to Q$ and $\bar{G} : Q \times U \to Y$ such that*

$$\pi(F(s, u)) = \bar{F}(\pi(s), u), \qquad G(s, u) = \bar{G}(\pi(s), u) \quad (s \in S, u \in U). \tag{1}$$

We denote the tuple by $\mathsf{TAA}_G(M) = (Q, \pi, \bar{F}, \bar{G})$. Its printed operator name is ⊂., pronounced TAA.

The subscript G is necessary. A contraction can preserve one observation while losing information required for another.

Theorem 1 (Trace preservation)**.** *For each initial state s_0 and finite input word, the system M and its contraction give equal observations at every step when the retained initial state is $\pi(s_0)$.*

Proof. Assume $q_k = \pi(s_k)$. The first identity in (1) gives $q_{k+1} = \bar{F}(q_k, u_k) = \pi(F(s_k, u_k)) = \pi(s_{k+1})$. The second identity gives $\bar{G}(q_k, u_k) = G(s_k, u_k)$. Since $q_0 = \pi(s_0)$, induction proves both statements for each step. □

For relations with multiple witnesses, preservation of the set of output values does not prove preservation of witness multiplicity. A claim of unique witness requires a bijection of the applicable witness fibers or an independent exact count. A claim of unique output requires equality of the applicable output sets.

2 Retained affine cost

An interval A carries the state transition $T_A(x) = F_A x + b_A$ and the cost $J_A(x) = x^{\mathsf{T}} Q_A x + q_A^{\mathsf{T}} x + d_A$, with Q_A symmetric. For interval B after A, define $T_{BA} = T_B \circ T_A$ and $J_{BA} = J_A + J_B \circ T_A$.

Proposition 1 (Quadratic energy composition)**.** *The composed coefficients are*

$$\begin{aligned} F_{BA} &= F_B F_A, & b_{BA} &= F_B b_A + b_B, \\ Q_{BA} &= Q_A + F_A^{\mathsf{T}} Q_B F_A, \\ q_{BA} &= q_A + F_A^{\mathsf{T}}(q_B + 2Q_B b_A), \\ d_{BA} &= d_A + d_B + q_B^{\mathsf{T}} b_A + b_A^{\mathsf{T}} Q_B b_A. \end{aligned}$$

This composition is associative and gives the exact chronological cost for each input state.

Proof. Substitute $F_A x + b_A$ into J_B and collect quadratic, linear, and constant coefficients. The displayed formulas result. For a third interval C, both groupings give the transition $T_C \circ T_B \circ T_A$ and cost $J_A + J_B \circ T_A + J_C \circ T_B \circ T_A$. Therefore both groupings agree as functions. □

Example. Let $T_A(x) = 2x + 1$, $J_A(x) = x^2 + 1$, $T_B(y) = 3y - 2$, and $J_B(y) = 2y^2 + 3y + 4$. Then $T_{BA}(x) = 6x + 1$ and $J_{BA}(x) = 9x^2 + 14x + 10$. At $x = 2$, the direct costs are 5 and 69, giving 74. The retained polynomial also gives 74. The companion verifier checks the exact rational identities and a family of associative compositions.

3 Certificates and meaning

A C. certificate states the artifact digest, claim, admissible input domain, observation contract, checker version, verdict, and evidence digest. The signature authenticates the issuer's statement; the independent checker establishes whether the evidence supports the declared claim. A transparency receipt records registration under one service's policy. None of these substitutes for the others.

The word *exact* is relative to the declared observation and domain. The definition gives no claim that arbitrary systems admit a small retained state or that arbitrary program properties are efficiently decidable.

4 Symbol and prior results

The symbol C. consists of U+1455 CANADIAN SYLLABICS TA and U+002E FULL STOP. The existing glyph has a linguistic identity in the Unified Canadian Aboriginal Syllabics block. This paper assigns a mathematical use to those characters; it does not claim to create them. The software identifier is `taa`, with the versioned name `taa.exact-state-contraction.v1`.

The author's earlier work proved retained affine-cost identities, minimal linear recovery ranks, and exact coefficient certificates for specific contracts [1]. Established quotient structures, bisimulation, automata minimization, and affine composition remain the mathematical background of this proposal.

References

[1] lmtlssss, *Affine Cycles: Exact Cut Defects and Minimal Retained State*, Zenodo (2026), DOI: 10.5281/zenodo.23931722.

[2] Unicode Consortium, *Unified Canadian Aboriginal Syllabics*, Unicode Character Database, U+1455.

[3] IETF, *An Architecture for Trustworthy and Transparent Digital Supply Chains*, RFC 9943 (2026).